

# Soumyadeep Nath

Data Scientist & AI Builder | Python · SQL · R · ML · GenAI · Full-Stack AI

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## PROFESSIONAL SUMMARY

From literature to data — building the bridge deliberately. Currently completing an Executive PG Programme in Data Science & AI at IIT Roorkee, working hands-on with Python, SQL, Machine Learning, and Generative AI systems including RAG pipelines, embeddings, and LLM APIs. Built and deployed multiple live AI applications spanning full-stack GenAI platforms, data analytics, and backend automation — including **AlignResume**, **Future Fit**, and **Overture Outreach**. A background in academic writing and language analysis provides a unique edge: translating complex model outputs into clear, compelling narratives that stakeholders understand.

## EDUCATION & TRAINING

### Executive PG Programme in Data Science & AI – In Progress

Feb 2026 – Present

Indian Institute of Technology (IIT) Roorkee | Remote

Coursework: Python · SQL · NumPy · Pandas · Machine Learning · Statistics · Data Visualization · Generative AI · Prompt Engineering · RAG · LLM APIs · Embeddings · Vector Databases

- Engineered data pipelines and automated ETL workflows using Python (Pandas, NumPy) to clean, transform, and prepare structured and unstructured datasets for downstream analysis.
- Queried relational databases using advanced SQL — window functions, joins, subqueries, and aggregations — extracting actionable, analysis-ready business insights.
- Constructed EDA frameworks applying statistical reasoning, hypothesis testing, and A/B testing methodology to validate data-driven decisions.
- Developed and evaluated supervised Machine Learning models — regression, classification, and tree-based ensembles (Random Forest, Gradient Boosting) — using scikit-learn.
- Implemented Generative AI applications by integrating LLM APIs, engineering structured prompts, and building RAG systems with embeddings and FAISS/Chroma vector databases.
- Translated complex model outputs into structured data narratives and interactive dashboards using Matplotlib, Seaborn, and Plotly; documented findings in reproducible Jupyter notebooks.

### M.A. in English Literature — First Division

Oct 2022 – Jul 2024

Presidency University, Kolkata

### B.A. Honours in English Literature — First Division

Sep 2019 – Jul 2022

Presidency University, Kolkata

## PROJECTS

### AlignResume — AI-Powered Resume Optimization Platform

May 2026

Live: [align-resume-beta.vercel.app](https://align-resume-beta.vercel.app) · GitHub: [github.com/sdn9300/align-resume](https://github.com/sdn9300/align-resume) · Stack: Next.js · TypeScript · Tailwind CSS · Groq API (LLaMA 3.3 70B) · Playwright · Zod · Vercel

- Architected and deployed an end-to-end full-stack AI web application that parses resumes and job descriptions, scores alignment quality, and generates ATS-optimized bullet rewrites via a structured JSON prompt pipeline over the Groq API — with Zod schema validation ensuring deterministic, parseable LLM outputs.
- Implemented an automated truthfulness guardrail layer that validates rewritten bullets against the original resume, blocking export if hallucinated metrics, employers, or skills are detected — prioritizing honesty over inflation.
- Engineered Playwright-based headless PDF export, rate-limited API proxy (10 runs/hr per IP), and GitHub-Vercel CI/CD auto-deploy; deepened expertise in LLM application architecture, prompt engineering, and production-grade full-stack AI deployment.

### Future Fit — AI-Powered Skill Trend Intelligence Platform

Jun 2026

Live: [skilltrendanalysis-rf3zefgsjaa4l9f8pu2prb.streamlit.app](https://skilltrendanalysis-rf3zefgsjaa4l9f8pu2prb.streamlit.app) · GitHub: [github.com/sdn9300/Future-Fit-AI-Powered-Skill-Trend-Analysis](https://github.com/sdn9300/Future-Fit-AI-Powered-Skill-Trend-Analysis)

Stack: Python · Pandas · Plotly · Streamlit · Prophet · mlxtend (Apriori) · SciPy · Groq API (LLaMA 3.3 70B)

- Built an end-to-end Time Series and NLP analysis platform processing 224,605 skill mentions extracted from 50,000+ global AI/Data Science job postings (2020–2026), using a taxonomy-driven regex extraction engine with word-boundary matching across 100+ skills.

- Engineered a 10-step deterministic cleaning pipeline and developed a Market Basket Analysis (Apriori) engine mapping 230 frequent skill itemsets and extracting association rules (lift  $\geq 0.90$ , confidence  $\geq 0.38$ ) to identify strong co-occurrence trends like PyTorch  $\rightarrow$  TensorFlow.
- Implemented a tiered yearly forecasting model using Prophet to project demand shares into 2027, integrating a dynamic fallback to linear trend extrapolation if the 80% confidence interval width exceeded a  $1.5\times$  variance threshold.
- Deployed a 6-panel Streamlit dashboard featuring Plotly charts, interactive association rules, and a Groq LLM-powered Skill Gap Advisor enriched with association rule insights to generate contextual learning paths.

### Gleaner — Multi-Board Job Scraper & ETL Pipeline

Jun 2026

GitHub: [github.com/sdn9300/gleaner-job-scout](https://github.com/sdn9300/gleaner-job-scout)

Stack: Python · BeautifulSoup4 · Firecrawl API · Google Sheets API · YAML · Git

- Architected and implemented a modular ETL pipeline using the Adapter Design Pattern to concurrently fetch, normalize, and aggregate job listings from 4+ major platforms (Naukri, Indeed, RemoteOK, Wellfound/WWR).
- Engineered robust scraping mechanics using stealth configuration headers, custom bot user-agents, and Firecrawl API's structured extraction logic to ensure reliable, anti-bot-resistant data retrieval.
- Designed data processing stages utilizing exact and similarity-based deduplication algorithms, role-relevance keyword analysis, and location-based filters to clean raw unstructured payloads.
- Integrated automated reporting channels by establishing programmatic write pathways to local CSV outputs and secure synchronization with cloud-hosted spreadsheets using the Google Sheets API (gsread / google-auth).
- Developed a configurable CLI engine driven by centralized YAML files, allowing granular execution control over logging levels, output size limits, and targeted source execution.

### Overture Outreach — AI-Powered Cold Email Outreach Platform

Jun 2026

Live: [frontend-three-azure-97.vercel.app](https://frontend-three-azure-97.vercel.app) · GitHub: [github.com/sdn9300/Overture-Outreach](https://github.com/sdn9300/Overture-Outreach)

Stack: Next.js · TypeScript · Python · Gmail OAuth2 API · Groq API (LLaMA 3.3 70B) · Prometheus · SQLite · Tailwind CSS · Vercel

- Architected and implemented a modular backend outreach engine using Python that orchestrates a dual-stage execution pipeline — deterministic HTML formatting followed by dynamic tone polishing via the Groq LLM API.
- Configured secure Google OAuth2 flow handling using Google Client APIs to obtain authorization tokens, enabling programmatic, rate-safe cold mail delivery directly through the Gmail API.
- Implemented asynchronous background execution with a Redis queue system and instrumentation via a Prometheus metrics server to expose live analytics counters on active mailings.
- Designed a responsive Next.js frontend dashboard using TypeScript and Tailwind CSS, deployed on Vercel, to display SQLite run histories, metrics, contact table filters, and live mail preview/composition.

## SKILLS

|                     |                                                                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages           | Python, SQL, TypeScript, R                                                                                                                                     |
| Frameworks & Tools  | Next.js, Tailwind CSS, Streamlit, Vercel, Playwright, Git / GitHub, Firecrawl API, YAML, Gmail OAuth2 API, Prometheus                                          |
| Libraries           | NumPy, Pandas, Matplotlib, Seaborn, Plotly, scikit-learn, SciPy, Prophet, mlxtend, Zod, BeautifulSoup4, gsread                                                 |
| Databases           | SQLite, Redis                                                                                                                                                  |
| Machine Learning    | Regression, Classification, Random Forest, Gradient Boosting, Model Evaluation, Time Series Forecasting (Prophet), Association Rule Mining (Apriori / mlxtend) |
| Generative AI       | LLM APIs (Groq), Prompt Engineering, Structured JSON Pipelines, RAG, Embeddings, Vector Databases (FAISS / Chroma), Guardrail Design                           |
| Data Analytics      | EDA, Data Cleaning, ETL Pipelines, Web Scraping, Statistical Analysis, Hypothesis Testing, A/B Testing, Data Storytelling                                      |
| Visualization       | Matplotlib, Seaborn, Plotly, Jupyter Notebook                                                                                                                  |
| Professional Skills | Critical Thinking, Analytical Reasoning, Communication, Attention to Detail, Adaptability                                                                      |

## CERTIFICATIONS

### Machine Learning Statistical Foundations Professional Certificate

LinkedIn Learning × Wolfram Research | Completed: Mar 2026 | Skills: Machine Learning · Linear Algebra · Wolfram Language

### Anaconda Python for Data Science Professional Certificate

LinkedIn Learning × Anaconda | Completed: Mar 2026 | Skills: Python · Data Analysis · Statistics

### Career Essentials in Generative AI

LinkedIn Learning × Microsoft | Completed: Mar 2026 | Skills: Generative AI · Microsoft Copilot · Responsible AI